

Github: <https://github.com/mrukudcode/Genai>

Task	Model	Classification (Success/Failure)	Observation (What actually happened?)	Why did this happen? (Architectural Reason)
Generation	BERT	Failure	Pipeline fails or produces meaningless output. In many cases, text-generation is not supported.	BERT is an encoder-only model. It is trained to understand complete input sequences, not to autoregressively generate the next token.
	RoBERTa	Failure	Similar to BERT: cannot generate coherent text or fails at initialization.	RoBERTa is also encoder-only, optimized for understanding tasks. It lacks a decoder and causal language modeling objective.
	BART	Success	Generates coherent continuation of the prompt. Output stays mostly on topic.	BART is an encoder–decoder (seq2seq) model. The decoder is trained to generate tokens sequentially, making it suitable for text generation.
Fill-Mask	BERT	Success	Correctly predicts words like “create”, “generate”.	BERT is explicitly trained using Masked Language Modeling (MLM), where masked tokens are predicted

				using surrounding context.
	RoBERTa	Success	High-quality predictions, often more confident than BERT.	RoBERTa improves upon BERT's MLM training with more data, dynamic masking, and better optimization.
	BART	Partial Success	Can predict a word, but results may be less precise than BERT/RoBERTa.	BART supports denoising objectives, but MLM is not its primary training task, so it is less specialized for fill-mask.
Q and A	BERT	Partial / Poor	Produces vague or incorrect answers when using base model.	The base BERT model is not fine-tuned for extractive QA (SQuAD), so it lacks task-specific alignment.
	RoBERTa	Partial / Poor	Slightly better than BERT but still unreliable.	Despite stronger pretraining, RoBERTa still requires QA fine-tuning to perform accurately.
	BART	Failure / Weak	Often fails to extract the correct span or gives irrelevant output.	BART is a generative seq2seq model, not optimized for extractive QA without task-specific fine-tuning.

The objective of this assignment is to analyze and compare the architectural differences between transformer-based models like BERT, RoBERTa, and BART by evaluating their performance across multiple NLP tasks. By deliberately applying these models to tasks beyond their primary design, the assignment aims to highlight how model architecture (encoder-only vs. encoder-decoder) and task-specific training influence model behavior, strengths, and limitations in real-world applications.

Through this assignment, I gained a clear understanding of how transformer model architecture directly affects task performance. By benchmarking BERT, RoBERTa, and BART across text generation, masked language modeling, and question answering, I observed that:

- Encoder-only models like BERT and RoBERTa perform well on understanding-based tasks such as masked word prediction but struggle with text generation, as they are not designed to generate sequences autoregressively.
- Encoder-decoder models like BART are better suited for generation and sequence-to-sequence tasks, demonstrating more coherent outputs when compared to encoder-only models.
- Using base (non-task-fine-tuned) models for question answering resulted in poor or random answers, highlighting the importance of task-specific fine-tuning.
- Model warnings and low confidence scores helped reinforce the concept that pretraining objectives strongly influence downstream task performance.
- This assignment improved my practical understanding of why model selection matters in real-world Generative AI applications, rather than relying on a one-model-fits-all approach.

Overall, the hands-on experiments helped bridge the gap between theoretical knowledge of transformers and practical model behaviour.

Frameworks/Libraries:

- Hugging Face transformers

- Hugging Face pipeline

Models Used:

- bert-base-uncased
- roberta-base
- facebook/bart-base